Vaibhav Hiwase



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AI/ML Engineer with 7+ years of experience building intelligent document processing systems and scalable ML pipelines. Skilled in Python, PyTorch, LLMs, and cloud-native architectures. Specialized in multimodal information extraction, LLM prompt engineering, and GenAI workflows using tools like MCP and Langchain. Proven success improving system accuracy and latency while driving ML solutions into production at scale.

Education

AUG 2016 – AUG 2018

Master of Technology in Computer Science and Engineering/RCOEM, Nagpur, India

AUG 2012 – AUG 2016

Bachelor of Engineering in Information Technology/SVPCET, Nagpur, India

Technical Skills

- **Languages:** Python (Advanced), Bash (Basic)
- **ML/LLM:** PyTorch, Scikit-learn, Azure Custom Vision, Llama2, FLAN-T5, Sentence Transformers, Prompt Engineering, FAISS
- **LLMs:** OpenAI, Llama, Ollama , FLAN-T5, Sentence Transformers, LLM Agents, Model Context Protocol (MCP)
- **MLOps & Cloud**: Databricks , Azure (AKS, ML Studio, Functions, Key Vault, Azure API Management, Function Apps, Azure container apps), Docker, Kubernetes, GitHub Actions, OAuth2.0

Inferencing:

- **Databases**: PostgreSQL, MySQL, Redis
- **Observability & Monitoring:** Arize Phoenix, RQ Dashboard, Opik by Comet, Power BI
- **Tools and Libraries:** TorchServe, VLLM, Ollama, FAISS, Langchain, LlamaIndex, Git, Jupyter, Spyder, VS Code, Cursor

Experience

SEPT 2023 – PRESENT (REMOTE)

Technical Lead ML & AI /TechnoMile, India

- Built a multimodal CLIN extraction system using Azure Custom Vision and LLMs, improving accuracy from 58% → 90% across various contract types to build contract intelligence, and competitive insights, specifically tailored for the Government Contracting, Aerospace & Defense sectors.
- Refactored pipeline into modular services (CLIN, Clause, Section, Form), enabling scalable orchestration via AWS Step Functions
- Implemented Model Context Protocol (MCP) to connect LLMs securely to live systems; built agents to query APIs and databases using natural language.
- Created a tool that visually links AI-extracted data back to its source in the PDF contract using bounding boxes. This helped users easily verify outputs, reduce errors, and catch unsupported AI-generated fields.
- Integrated LLM observability with Arize Phoenix to trace prompts, detect anomalies, and debug token-level errors.
- Reduced processing time by 40% via chunk-level parallelism and optimized PDF-to-image generation with a memory-efficient pipeline.

SEPT 2021 – AUG 2023 (REMOTE) (2 YEARS 1 MONTH)

Senior Consultant/Celebal Technologies, India

- Led the development of a proprietary technique for spotting payslip forgery with an 85.23% accuracy.
- Designed and implemented a high-performance architecture by leveraging technologies such as Redis, Kubernetes, API Management, Key Vault, docker, AKS and OAuth2.0, resulting in a 10x reduction in processing time.
- Worked in a collaborative team environment in partnership with Microsoft to develop a forecasting platform
- Mentored and directed team in various areas such as key information extraction, document similarity algorithms, Hinglish language model creation, and supervised asset for counting exercise reps using video and camera from mobile and IoT device.

AUG 2018 TO AUG 2021 (3 YEARS 1 MONTH)

Machine Learning Engineer/Konverge.AI, India

- Built Smart Content Extraction pipeline with clustering for paragraph and table detection in scanned documents.
- Developed APIs for document ingestion using Airbyte and WEAV Core, improving pipeline robustness and modular.
- Trained Mask-RCNN for pixel-level classification of pathogens in medical images; delivered deployable diagnostic tools.
- Created GIS system to extract location data from scanned government documents for spatial analysis.

Projects

Multimodal AI Systems & Prompt Engineering

- Improved document data extraction accuracy by 32% using vision + language models across 5 domains.
- Refactored a legacy monolithic pipeline into modular services (FORM, CLIN, SECTION, CLAUSE IE), enabling reusable ML pipeline library (training, inference, deployment) compatible with TorchServe, VLLM, Ollama.
- Built layout-aware CLIN detection model with Azure Custom Vision; scaled from 94% to 98% accuracy using feedback loop on 264 PDFs.
- Developed JSON alignment and confidence scoring system using SequenceMatcher for human-in-the-loop validation.
- Reduced CLIN processing latency by ~40% (370s → 220s) through chunk-level parallelism and memory optimization for concurrent document segment processing.

Large Language Model Training

- Fine-tuned Spanish LLM (Sentence Transformer) on Walmart Mexico catalog for ad keyword generation..
- Deployed FLAN-T5 and BiVAE models for semantic retrieval and collaborative filtering; integrated with FAISS (Facebook AI Similarity Search) to improve query latency.

Research Collaboration - Forecasting Platform

- Integrated training loss monitoring into several forecasting models including DeepMC, TFT, DartsNBEATS, and DartsTCN through joint efforts.
- Collaborated to design a forecasting platform that enhanced decision-making in energy portfolio management by optimizing renewables monetization and orchestration. Streamlined pipeline execution using Azure Machine Learning Studio and Azure Functions.

Payslip Forgery Detection

- Led end-to-end development—from research to infrastructure setup—for a payroll management startup, including architecture, engineering, and monitoring.
- Developed a proprietary technique to detect payslip forgery with 85.23% accuracy, later enhanced with metadata and pixel-level analysis, achieving document classification accuracies of 90.87% (payslip) and 95.72% (blur detection).
- Cut document processing time by 10x using Redis queues and Dockerized Flask workers. Collaborated with team to deployed the application to Kubernetes with KEDA-based auto-scaling triggered by Redis queue size, ensuring scalability under load.
- Integrated RQ Dashboard for async task processing of Redis Queue, improving visibility into job execution and worker health.
- Secured communication between the Flask backend and React frontend using OAuth 2.0 and OpenID Connect and Azure API management.
- Streamlined CI/CD by using GitHub Actions and building a production-ready multi-stage container that unified Flask and React apps, reducing deployment complexity and cost via Azure Container Apps.

Certifications

Google Project Management (In progress)
STATSX0001: Statistical Learning by StanfordOnline
Oauth2.0
Docker
Data Science

Participations

Participated in Microsoft's AI/ML competition (2022)
Artificial Intelligence and Machine Learning Workathon (2021)
Organized and led CSI events (2013-2016)
National Cadet Corps (NCC) cadet with 'A' certificate (2008-2009)